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⑤ Registers and counters – Assignment

1.> Design a counter with T flip-flops whose count sequence is 0, 2, 4, 5, 3, 1, 0. . . .

= Let

current state			next state			required input		
Q_2	Q_1	Q_0	Q_2	Q_1	Q_0	T_2	T_1	T_0
0	0	0	0	1	0	0	1	0
0	0	1	0	0	0	0	0	1
0	1	0	1	0	0	1	1	0
0	1	1	0	0	1	0	1	0
1	0	0	1	0	1	0	0	1
1	0	1	0	1	1	1	1	0
1	1	0	x	x	x	x	x	x
1	1	1	x	x	x	x	x	x

$T_0 =$

Q_2 \ Q_1Q_0	00	01	11	10
0	0	1	0	0
1	1	0	X	X

$T_2 = Q_2\bar{Q}_0 + \bar{Q}_2\bar{Q}_1Q_0$

$T_1 =$

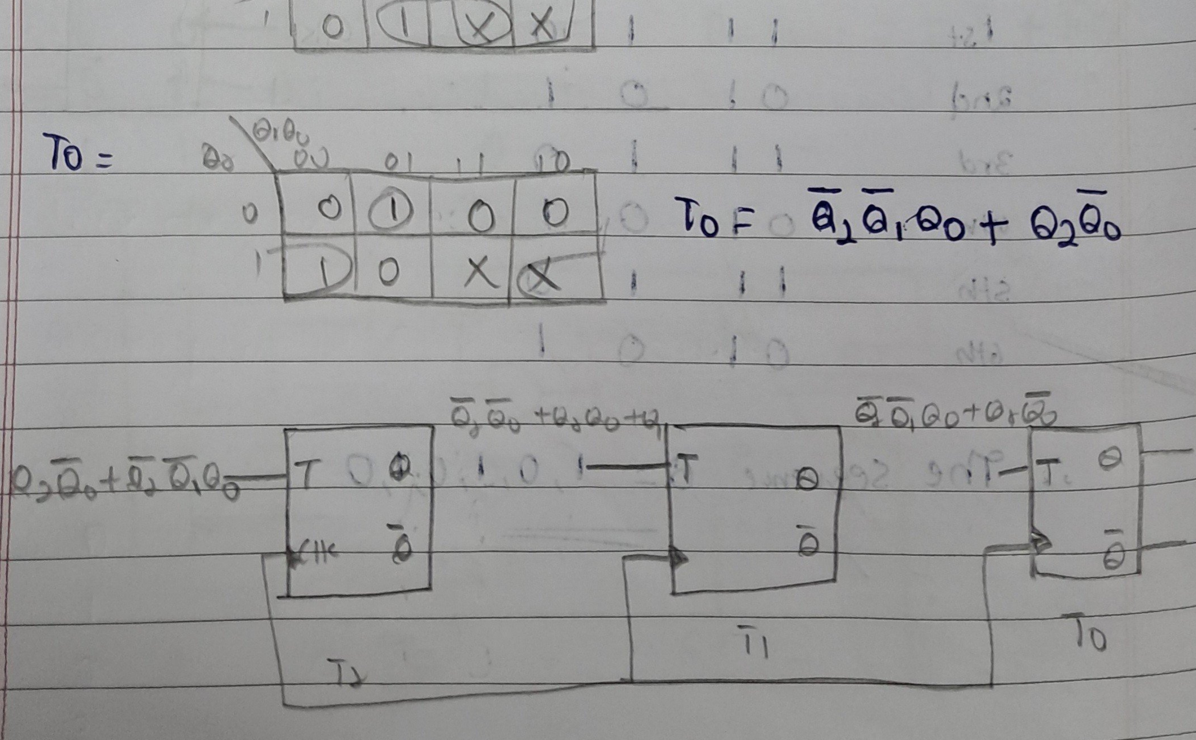
Q_2 \ Q_1Q_0	00	01	11	10
0	1	0	1	1
1	0	1	X	X

$T_1 = \bar{Q}_2\bar{Q}_0 + Q_2Q_0 + Q_1$

$T_0 =$

Q_2 \ Q_1Q_0	00	01	11	10
0	0	1	0	0
1	1	0	X	X

$T_0 = \bar{Q}_2\bar{Q}_1Q_0 + Q_2\bar{Q}_0$



2) Design a 3 bit up/down counter using negative edge triggered T-FF

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Input	Present state			next state			T flip-flop input		
M	Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	T_2	T_1	T_0
0	0	0	0	0	0	1	0	0	1
0	0	0	1	0	1	0	0	1	1
0	0	1	0	0	1	1	0	0	1
0	0	1	1	1	0	0	1	1	1
0	1	0	0	1	0	1	0	0	1
0	1	0	1	1	1	0	0	1	1
0	1	1	0	1	1	1	0	0	1
0	1	1	1	0	0	0	1	1	1
1	0	0	0	1	1	1	1	1	1
1	0	0	1	0	0	0	0	0	1
1	0	1	0	0	0	1	0	1	1
1	0	1	1	0	1	0	0	0	1
1	1	0	0	0	1	1	1	1	1
1	1	0	1	1	0	0	0	0	1
1	1	1	0	1	0	1	0	1	1
1	1	1	1	1	1	0	0	0	1

$\therefore T_0 = 1$

$T_1 =$

$MQ_2 \backslash Q_1Q_0$	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	1	0	0	1
10	1	0	0	1

$\therefore T_1 = \bar{M} Q_0 + M \bar{Q_0}$

$T_2 =$

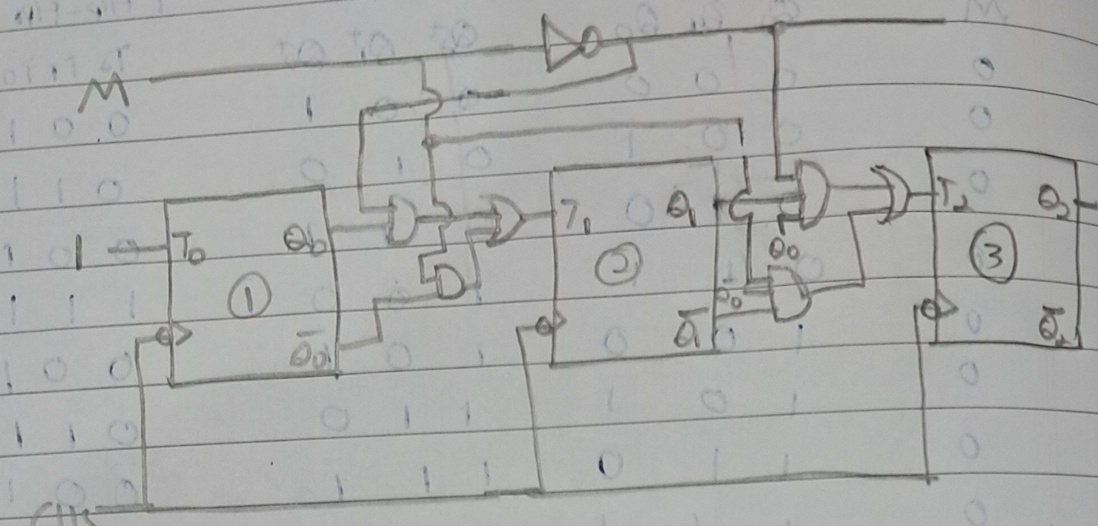
$MQ_2 \backslash Q_1Q_0$	00	01	11	10
00	0	0	1	0
01	0	0	1	0
11	1	0	0	0
10	1	0	0	0

$\therefore T_2 = \bar{M} Q_1 Q_0 + M \bar{Q_1} \bar{Q_0}$

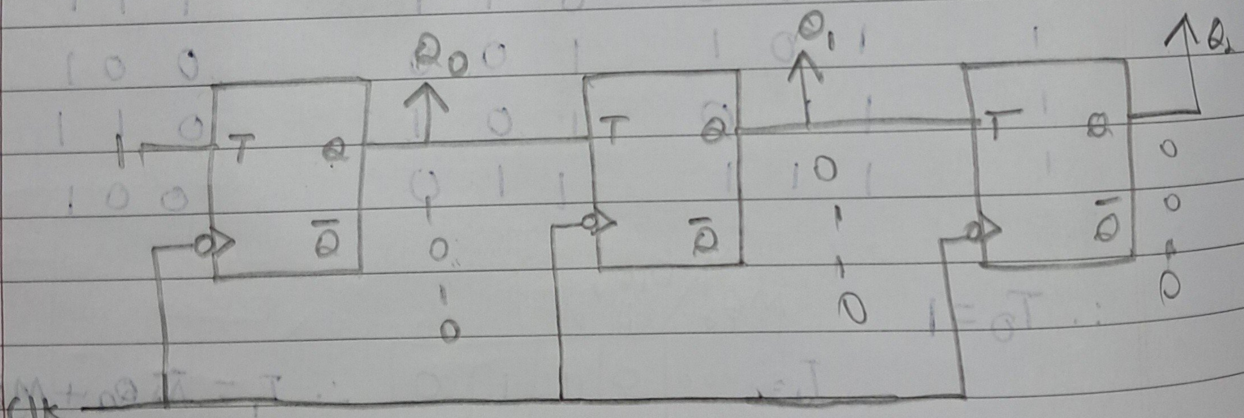
$\therefore \quad T_0 = 1$

$T_1 = \bar{M} Q_0 + M \bar{Q}_0$

$T_2 = \bar{M} Q_1 Q_0 + M \bar{Q}_1 \bar{Q}_0$



3.) The circuit in the figure below is a counter. What is the sequence that this circuit counts in.



= These are negative edge triggered T-flip-flops

assuming initial state $Q_0 Q_1 Q_2 = 000$.

Sequence	Q_2	Q_1	Q_0
0	0	0	0
1	0	0	1
2	0	1	0
3	1	1	1

4. 0 0 0

5 0 0 1

6 0 1 0

⋮ ⋮

∴ The sequence it follows is 000 → 001 → 010 → 111 → 000 ..